

at the edge greatly. “The shift to small language models in edge computing will improve inference in various industries, including automotive, retail, and advanced robotics. It also can train robots to react in virtually any scenario, enhancing their adaptability and performance across different clinical situations,” says the Nvidia report.

It is also expected that AI, together with virtual reality (VR), will help to train robots to handle complex tasks such as surgical assistance. Virtual worlds will also help to train the robots that interact with humans, by simulating different scenarios. This will reduce the cost of training and the risk of deployment, paving the way for greater adoption of co-working robots.

A recent MarketsandMarkets report projects the growth of the humanoid robot market from \$2.03 billion in 2024 to \$13.25 billion in 2029, with a compound annual growth rate (CAGR) of 45.5%. This growth will likely be driven by advances in AI and ML, improvements in humanoid robots, and the growing use of humanoids as educational robots. High initial investment and safety concerns need to be addressed to catalyse the growth in 2025 further.

Sustainability and green initiatives

The amount of computing power required for AI implementations, the data centres populating the cloud, and the uncountable number of mobile devices at the edge—the carbon footprint of all this might even be a bit too much for AI to comprehend. Still, deep within our hearts, we all know that we are heading towards destruction! The environmental impact does not end with the energy consumed by these, but also the resources consumed and the pollution caused during their production. Understandably, the tech



AI-powered robots are expected to make their presence felt in the physical world

industry and its large enterprise consumers are under great pressure from global agencies and governments to reduce and reverse their environmental impacts.

The industry is once again turning to its trusted friend—technology itself—to find ways to reduce energy consumption and improve sustainability by optimising architectures, hardware, code, and algorithms. They are also turning to renewable energy sources like solar energy and opting for more eco-friendly and sustainable manufacturing methods and packaging.

Apart from the actual tech processes, companies also need to optimise the carbon footprint of running the organisation per se, like transport and energy consumption on campus. So, companies have started using electric vehicles (EVs) in their fleet, and incentivising eco-friendly activities like using the stairs, avoiding plastic containers, and so on. While implementing these will be expensive initially, companies can no longer escape the regulatory requirements. That said, there will be significant cost savings in the long run.

Almost all industry majors, including Apple, Google, and Microsoft, have committed to becoming

carbon neutral or negative by 2030. Carbon neutrality requires changes across the entire business—from manufacturing processes and supply chain to product lifecycle and business operations—to ensure zero climate impact.

So how does one become carbon-negative? It means you do stuff to reverse environmental damage, like planting trees, generating enough renewable energy to feed back to the grid, capturing greenhouse gases from the environment, promoting the maintenance of rangelands, which sequester carbon dioxide in the soil and biomass, landfill management, and so on.

Companies can also gain carbon credits by funding organisations that engage in such activities; it is believed that this can be a means to uplift tribal populations. Companies pay them handsomely to do what comes naturally to them—fostering the environment. Companies earn carbon credits, while the tribes earn their sustenance with due respect!

And some more...

- **Spatial computing.** Spatial computing is a paradigm that digitally enhances the physical world. An amalgam of various technologies